CIS 6270 Practice Exam 2 for Exam 2 SOLUTIONS

Professor Pranam Chatterjee
TAs: Sophie Vincoff, Tong Chen, and Rosie Zhang

Fall 2026

Use the following conditions for a timed practice attempt:

- Please write your name on each page. Allow yourself 90 minutes.
- Work through the exam on your own before consulting the solutions. Write down any questions you would like to discuss with the instructor or TAs.
- Each part has six choices. Use the space provided to work through the problem, then box one answer letter. Only the boxed letter counts toward your score.
- Each labeled part is worth 0.5 points. We award full credit for a correct answer and zero otherwise.
- You may use one side of an 8.5in $\times$ 11in cheat sheet containing only material from this class. Keep your signed sheet with your completed work. Do not use a calculator, phone, or other device during the timed attempt.

The University of Pennsylvania is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. I will use this practice exam to assess my own understanding. I will complete the timed attempt using my own work before consulting the solutions or discussing answers.

This practice exam contains 12 points and does not contribute to your course grade.

Name: _____

Signature: _____

This page is left intentionally blank for scratch work

This practice exam is worth 12 points.

Fewer Sampling Steps with Class-Rate Flow Matching

You lead a group that designs peptide binders. An autoregressive model generates one residue per network call, so a 24-residue peptide takes 24 calls. With discrete flow matching, one call predicts residues at every position in parallel. To generate the same peptide in fewer calls, we would therefore need fewer than 24 sampling steps. For comparison, discrete flow matching samplers have used 1024 steps for text, 100 for images, and up to 10,000 for code.

What happens if we try to generate the entire peptide in one step? Each position is drawn independently given the same input, so a scaffold residue cannot depend on an anchor chosen in that step. In this exam, you will develop class-rate flow matching to reveal the anchors that carry the binding motif earlier on average. When we generate the remaining scaffold residues, we can then condition on the anchors already in place. During training, we still corrupt every position at randomly sampled times.

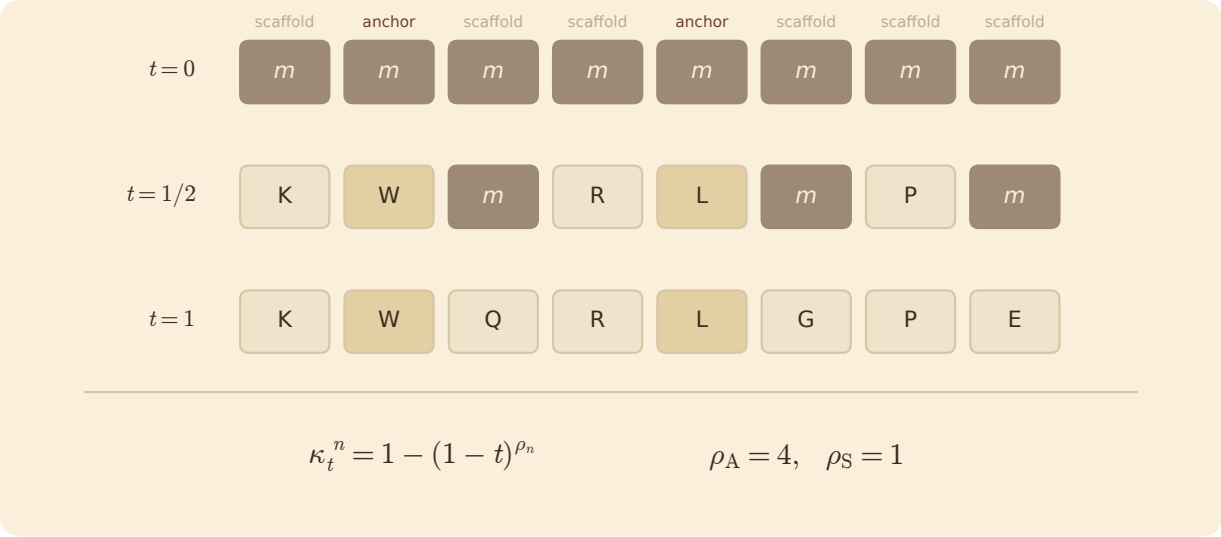


Figure 1. Class-rate flow matching on a peptide. We start with an all-mask peptide at $t = 0$ and generate a clean peptide at $t = 1$. A position of class n holds its data residue with probability $\kappa_t^n = 1 - (1 - t)^{\rho_n}$. We use a larger exponent ρ_n for anchor positions. The figure shows eight of the 24 positions. The row at $t = 1/2$ shows one sample from the path.

You will first use jump rates to describe how probability moves between peptide sequences. You will then choose a path for the anchor and scaffold residues, derive its velocity, and train a denoiser to sample from it. After examining how a corrector can revise a residue, you will consider a continuous flow on the simplex. Finally, you will guide generation toward binders with Doob's h-transform and interpret hypothetical results.

Research brief and notation

Each peptide $x = (x^1, \ldots, x^N)$ has $N = 24$ positions. Each position holds one of the 20 amino acids in $\mathcal{V}$, while $m \notin \mathcal{V}$ denotes the mask. We assign each position i to a class $c(i) \in \{\mathrm{A}, \mathrm{S}\}$. The six anchor positions carry the binding motif. The remaining 18 positions form the scaffold. In discrete flow matching, time runs from the source at $t = 0$ to the data at $t = 1$. We write the destination first, so $u_t(y, x)$ is the rate of jumping from x to y.

We start from an all-mask peptide and draw the clean peptide x_1 from the data distribution q. The probability of revealing a residue depends on its class,

$$\kappa_t^n = 1 - (1-t)^{\rho_n}, \qquad \rho_{\mathrm{A}} = 4, \qquad \rho_{\mathrm{S}} = 1,$$

so a position of class n holds its data residue with probability κ_t^n and remains masked with probability $1 - \kappa_t^n$. Setting $\rho_n = 1$ recovers the linear schedule $\kappa_t = t$. If we use this schedule for both classes, every position has the same reveal rate.

The denoiser $p^\theta_{1|t}(a \mid x)$ at position i is the probability that the data residue x_1^i is $a \in \mathcal{V}$, given the current peptide x. We write $x^{i \leftarrow a}$ for x with position i set to a, and δ for the indicator that its two arguments agree. Jumps change one position at a time, and $u_t^i(a, x)$ denotes the per-position velocity. For the parts on the simplex, we use the toy four-residue alphabet $\{\mathrm{L}, \mathrm{I}, \mathrm{K}, \mathrm{R}\}$. Logarithms are natural.

Step	**Points**
1. Read rates as velocities	2.0
2. Choose the class-rate path	2.0
3. Derive the velocity and the loss	2.0
4. Sample, correct, and check optimality	2.0
5. Relax the peptide to the simplex	2.0
6. Guide generation and read the results	2.0
Total: 24 parts	**12.0**

Step 1: Reading Rates as Velocities (2 points)

Before choosing a schedule for the anchors, we need to understand how jump rates change state probabilities. We can then check which dependencies we lose if we generate an entire peptide in one parallel step.

(a) We write each jump rate with the destination first. Which constraints must the rates satisfy for a fixed starting state?

A position currently holds the mask m, and its other states are the residues L, K, and R. Which vector $\big(u_t(\mathrm{L}, m),\ u_t(\mathrm{K}, m),\ u_t(\mathrm{R}, m),\ u_t(m, m)\big)$ is a valid probability velocity at this position? Choose ONE.

A. $(0.6,\ 0.9,\ 0.5,\ -1.0)$

B. $(0.6,\ -0.9,\ 0.5,\ -0.2)$

C. $(0.6,\ 0.9,\ 0.5,\ 2.0)$

D. $(0.2,\ 0.3,\ 0.5,\ 0)$

E. $(-0.6,\ -0.9,\ -0.5,\ 2.0)$

F. $(0.6,\ 0.9,\ 0.5,\ -2.0)$

(0.5 points)

For each starting state, we check the corresponding column of the velocity matrix. First, every off-diagonal entry must be nonnegative,

$$u_t(y, x) \geq 0 \quad \text{for } y \neq x,$$

and the full column for x must sum to zero:

$$\sum_y u_t(y, x) = 0, \qquad \text{so} \qquad u_t(x, x) = -\sum_{y \neq x} u_t(y, x).$$

Here the three rates out of the mask are 0.6, 0.9, and 0.5, so the diagonal entry is $-(0.6+0.9+0.5) = -2.0$. Therefore, Option F. The total rate of leaving the mask is 2.0, and the mean waiting time before this position is revealed is $1/2.0$.

In Option A, the off-diagonal rates are valid. Its entries sum to 1.0, so the velocity would fail to conserve probability. Option B has a negative rate toward K, which no jump process allows. Option C has a positive diagonal entry, so probability would flow out of every state at once. Option D is a probability vector over the four states, and its entries sum to one. Option E flips the sign of every entry, so the off-diagonal rates are negative and the diagonal is positive.

(b) Once we know the jump rates, we can use them to calculate how the probability of each state changes.

At one position, the probabilities of $(\mathrm{L}, \mathrm{K}, \mathrm{R}, m)$ are (0.1, 0.2, 0.3, 0.4). From the mask, the rates of revealing L, K, and R are $u_t(\mathrm{L}, m) = 1$, $u_t(\mathrm{K}, m) = 2$, and $u_t(\mathrm{R}, m) = 1$, respectively. All other off-diagonal rates are zero. Which vector gives $\big(\dot{p}_t(\mathrm{L}),\, \dot{p}_t(\mathrm{K}),\, \dot{p}_t(\mathrm{R}),\, \dot{p}_t(m)\big)$? Choose ONE.

A. $(0.4,\ 0.8,\ 0.4,\ -1.6)$

B. $(1,\ 2,\ 1,\ -4)$

C. $(0.1,\ 0.4,\ 0.3,\ -0.8)$

D. $(0.4,\ 0.8,\ 0.4,\ -0.4)$

E. $(0.04,\ 0.16,\ 0.12,\ -0.32)$

F. $(-0.4,\ -0.8,\ -0.4,\ 1.6)$

(0.5 points)

The master equation in velocity form collects the flux arriving at each state:

$$\dot{p}_t(y) = \sum_x u_t(y, x)\, p_t(x).$$

Only the mask has outgoing jumps, so the flux into each residue comes from the mask:

$$\dot{p}_t(\mathrm{L}) = 1 \cdot 0.4 = 0.4, \qquad \dot{p}_t(\mathrm{K}) = 2 \cdot 0.4 = 0.8, \qquad \dot{p}_t(\mathrm{R}) = 1 \cdot 0.4 = 0.4.$$

The mask loses all of it, $\dot{p}_t(m) = -(1 + 2 + 1)(0.4) = -1.6$. Therefore, Option A. The four rates of change sum to zero, so total probability stays at one.

Option B lists the rates themselves, without weighting them by the probability 0.4 of sitting at the mask. Option C weights each rate by the probability of the destination, although the flux leaves the mask. Option D keeps the arrivals and sets the departure to a single rate, so the entries no longer sum to zero. Option E multiplies by both probabilities, which counts the mask twice. Option F reverses every sign, which would drain the residues into the mask.

(c) Suppose we sample two residues in parallel from a single network prediction. Can this step reproduce a dependence between the two positions?

A two-residue peptide forms a salt bridge, so its data distribution places probability 1/2 on LK and 1/2 on KL. From the all-mask peptide, we take one step that draws both positions at once from their own marginals. What is the probability that this step produces a data peptide? Choose ONE.

A. 1

B. 3/4

C. 1/2

D. 1/4

E. 1/8

F. 0

(0.5 points)

With the all-mask source there is one source state, so the per-position marginals come from the data distribution alone. Position 1 holds L with probability 1/2 and K with probability 1/2, and position 2 does the same. A step that draws both at once from these marginals gives

$$\Pr(\text{LK}) = \Pr(\text{KL}) = \Pr(\text{LL}) = \Pr(\text{KK}) = \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4},$$

and two of those four are data peptides. Therefore, Option C. In the data, choosing L at one position fixes K at the other. Independent draws lose this dependence and produce two identical residues half the time.

Option A would hold if the positions were independent under the data distribution. Option B counts three of the four outcomes. Option D counts one outcome, although LK and KL both qualify. Option E would need four independent binary draws. Option F would hold if the marginals placed no mass on L and K, and each marginal places half its mass on each.

(d) To quantify the error in part (c), we compare the joint distribution with the product of its per-position marginals using KL divergence. Averaging this divergence over the source gives the conditional total correlation of the coupling.

For the data distribution in (c) with the all-mask source, what is the conditional total correlation? Choose ONE.

A. 0

B. $\frac{1}{2}\log 2$

C. $\log(4/3)$

D. $\log 2$

E. $\log 4$

F. $4\log 2$

(0.5 points)

The conditional total correlation compares the joint distribution of the data peptide given the source with the product of its per-position marginals. One source state gives one term:

$$\sum_{x_1} q(x_1)\log\frac{q(x_1)}{\prod_i q_i(x_1^i)} = 2\cdot\tfrac{1}{2}\log\frac{1/2}{(1/2)(1/2)} = \log 2.$$

Therefore, Option D. The joint distribution assigns probability to only two peptides, while the product of the marginals assigns equal probability to all four. Their KL divergence is $\log 2$ nats. Rectifying the coupling can reduce this error in a one-step sampler.

Option A would hold if the two positions were independent given the source. Option B halves the correct value. Option C is the divergence for a joint distribution that places 3/4 of its mass where the product places 1. Option E doubles the correct value, which would need a joint supported on one of the four peptides. Option F is the value for a sixteen-outcome version of the same comparison.

Step 2: Choosing the Class-Rate Path (2 points)

With a shared schedule, anchor and scaffold positions have the same probability of remaining masked. We now give each class its own schedule so that anchors become visible earlier on average. To construct the velocity, we first need the reveal rate implied by each schedule.

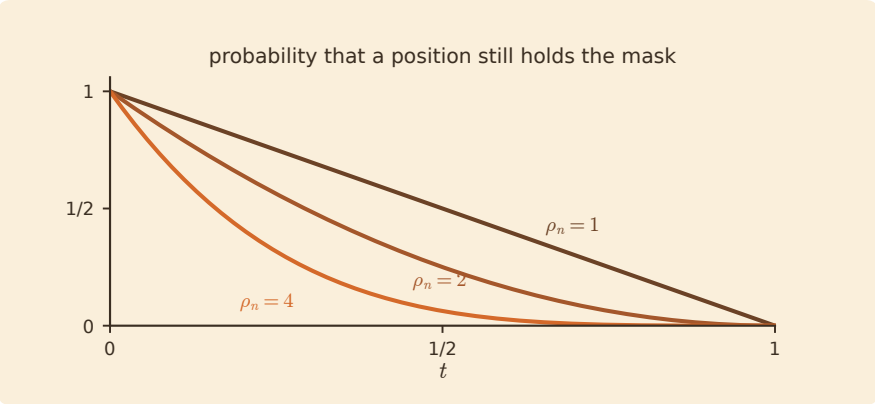


Figure 2. Probability that a position remains masked for three values of ρ_n. When $\rho_n = 1$, this probability decreases linearly with time.

(a) To find the reveal rate along a mixture path, we need the probability that a position is still masked and the rate at which that probability decreases.

With $\kappa_t^n = 1 - (1-t)^{\rho_n}$, what is the schedule factor $\dot{\kappa}_t^n/(1-\kappa_t^n)$? Choose ONE.

A. $\rho_n(1-t)^{\rho_n-1}$

B. $\dfrac{\rho_n}{1-t}$

C. $\dfrac{1}{1-t}$

D. $\dfrac{\rho_n}{t}$

E. $\dfrac{\rho_n(1-t)^{\rho_n-1}}{1-(1-t)^{\rho_n}}$

F. $\dfrac{\rho_n}{(1-t)^{\rho_n}}$

(0.5 points)

Differentiate the schedule:

$$\dot{\kappa}_t^n = \frac{\mathrm{d}}{\mathrm{d}t}\big(1-(1-t)^{\rho_n}\big) = \rho_n(1-t)^{\rho_n-1}.$$

The probability that the position still holds the mask is $1 - \kappa_t^n = (1-t)^{\rho_n}$, so

$$\frac{\dot{\kappa}_t^n}{1-\kappa_t^n} = \frac{\rho_n (1-t)^{\rho_n - 1}}{(1-t)^{\rho_n}} = \frac{\rho_n}{1-t}.$$

Therefore, Option B. The class exponent multiplies the linear schedule's reveal rate by ρ_n. At any given time, a masked anchor is therefore revealed at four times the rate of a masked scaffold position. Setting $\rho_n = 1$ recovers the linear schedule's rate, $1/(1-t)$.

Option A is $\dot{\kappa}_t^n$ without the division by $1 - \kappa_t^n$. Option C is the linear schedule's factor, which matches only when $\rho_n = 1$. Option D divides by t, which belongs to the discrete diffusion convention where $t = 0$ is the clean sequence. Option E is $\dot{\kappa}_t^n / \kappa_t^n$, the factor of the backward velocity we use in 4(c). Option F divides by the masking probability a second time.

(b) Now that we have the reveal rate, how does the probability of remaining masked change over time?

Let $p_t(m)$ be the probability that a position of class n still holds the mask. Which equation is the master equation at the mask state? Choose ONE.

A. $\dot{p}_t(m) = \dfrac{\rho_n}{1-t}\, p_t(m)$

B. $\dot{p}_t(m) = -\dfrac{\rho_n}{1-t}\big(1 - p_t(m)\big)$

C. $\dot{p}_t(m) = -\dfrac{\rho_n}{1-t}$

D. $\dot{p}_t(m) = -\dfrac{\dot{\kappa}_t^n}{\kappa_t^n}\, p_t(m)$

E. $\dot{p}_t(m) = \dfrac{\rho_n}{1-t}\big(1 - p_t(m)\big)$

F. $\dot{p}_t(m) = -\dfrac{\rho_n}{1-t}\, p_t(m)$

(0.5 points)

A masked position becomes visible at a total rate of $\rho_n/(1-t)$. Because the process never remasks a residue, the master equation at the mask contains only the outgoing flux:

$$\dot{p}_t(m) = -\frac{\rho_n}{1-t}\, p_t(m).$$

Therefore, Option F. The marginal $p_t(m) = (1-t)^{\rho_n}$ satisfies it, since

$$\frac{\mathrm{d}}{\mathrm{d}t}(1-t)^{\rho_n} = -\rho_n (1-t)^{\rho_n - 1} = -\frac{\rho_n}{1-t}(1-t)^{\rho_n}.$$

This survival equation gives the masking probability for each class from its reveal rate.

Option A reverses the sign, so the mask would gain probability. Option B weights the rate by the probability of the residues, although the flux leaves the mask and the mask holds $p_t(m)$. Option C drops the factor $p_t(m)$, so the mask would empty at a rate that does not depend on how much is left in it. Option D uses the backward factor $\dot{\kappa}_t^n / \kappa_t^n$ from 4(c). Option E combines the wrong sign with the wrong factor.

(c) To sample over a finite time interval in Step 4, we will need the probability that a position remains masked throughout that interval.

For $s < t$, what is the probability that a position of class n still holds the mask at time t, given that it holds the mask at time s? Choose ONE.

A. $\left(\frac{1-t}{1-s}\right)^{\rho_n}$

B. $(1-t)^{\rho_n}$

C. $\left(\frac{1-s}{1-t}\right)^{\rho_n}$

D. $\big(1-(t-s)\big)^{\rho_n}$

E. $\frac{(1-t)^{\rho_n}-(1-s)^{\rho_n}}{1-(1-s)^{\rho_n}}$

F. $1-\left(\frac{1-t}{1-s}\right)^{\rho_n}$

(0.5 points)

To still hold the mask at t, the position has to hold the mask at s and then keep it over the interval, so

$$(1-\kappa_t^n) = (1-\kappa_s^n)\,\Pr\big(x_t^i = m \mid x_s^i = m\big).$$

Dividing gives the interval probability

$$\Pr\big(x_t^i = m \mid x_s^i = m\big) = \frac{(1-t)^{\rho_n}}{(1-s)^{\rho_n}} = \left(\frac{1-t}{1-s}\right)^{\rho_n}.$$

Therefore, Option A. The interval probabilities over a grid of times multiply to $(1-t)^{\rho_n}$, which is what lets us take steps of any size and still land on the same path.

Option B is the unconditional probability of holding the mask at t, with no conditioning on s. Option C inverts the ratio and exceeds one for $s < t$. Option D uses the elapsed time in place of the two survival probabilities, and it disagrees with the schedule except at $s = 0$. Option E is a difference of survival probabilities divided by a masking probability. Option F is the complement, the chance that the position is revealed during the interval.

(d) Our main runs use $\rho_{\mathrm{A}} = 4$ at the anchors and $\rho_{\mathrm{S}} = 1$ at the scaffold.

At $t = 3/4$, which pair gives the probability that an anchor position still holds the mask and the probability that a scaffold position still holds the mask? Choose ONE.

A. $(1/16,\ 1/4)$

B. $(1/256,\ 1/4)$

C. $(1/64,\ 1/4)$

D. $(1/4,\ 1/256)$

E. $(255/256,\ 3/4)$

F. $(1/256,\ 3/4)$

(0.5 points)

The probability of still holding the mask is $(1-t)^{\rho_n}$. At $t = 3/4$,

$$\text{anchor: } \left(\tfrac{1}{4}\right)^4 = \tfrac{1}{256}, \qquad \text{scaffold: } \left(\tfrac{1}{4}\right)^1 = \tfrac{1}{4}.$$

Therefore, Option B. Three quarters of the way through generation, an anchor remains masked with probability one in 256. A scaffold position remains masked with probability one in four. By this time, we can usually condition the remaining scaffold predictions on the visible anchors.

Option A uses the exponent 2 at the anchors. Option C uses the exponent 3 at the anchors. Option D exchanges the two classes, which would reveal the scaffold first. Option E gives the probability that each position has already been revealed, $1-(1-t)^{\rho_n}$. Option F takes the complement at the scaffold alone.

Step 3: Deriving the Velocity and the Loss (2 points)

We have chosen how each position moves from the mask to its clean residue. We can now derive the jump rates that generate this path for a known clean peptide. During sampling, we only know the current peptide, so we will train a denoiser to predict the clean residues from it.

(a) When the clean peptide is known, the conditional velocity directs each masked position toward its clean residue. To obtain a velocity we can use during sampling, we average over possible clean residues using the probabilities predicted by the denoiser.

Write d for $p^{\theta}_{1|t}(a \mid x)$. Which pair gives the conditional velocity $u^i_t(a, z \mid x_1)$ from the current residue z at a position of class n, and the model velocity $u^i_t(a, x)$? Choose ONE.

A. $\left(\frac{\rho_n}{1-t}\,\delta(a, x^i_1),\quad \frac{\rho_n}{1-t}\,d\right)$

B. $\left(\frac{\rho_n}{1-t}\big[\delta(a, x^i_1) - \delta(a, z)\big],\quad \frac{\rho_n}{1-t}\big[\delta(a, x^i) - d\big]\right)$

C. $\left(\frac{\dot\kappa^n_t}{\kappa^n_t}\big[\delta(a, x^i_1) - \delta(a, z)\big],\quad \frac{\dot\kappa^n_t}{\kappa^n_t}\big[d - \delta(a, x^i)\big]\right)$

D. $\left(\frac{\rho_n}{1-t}\big[\delta(a, x^i_1) - \delta(a, z)\big],\quad \frac{\rho_n}{1-t}\big[d - \delta(a, x^i)\big]\right)$

E. $\left(\frac{1}{1-t}\big[\delta(a, x^i_1) - \delta(a, z)\big],\quad \frac{1}{1-t}\big[d - \delta(a, x^i)\big]\right)$

F. $\left(\rho_n(1-t)^{\rho_n - 1}\big[\delta(a, x^i_1) - \delta(a, z)\big],\quad \rho_n(1-t)^{\rho_n - 1}\big[d - \delta(a, x^i)\big]\right)$

(0.5 points)

Along the mixture path, a position holds its source residue with probability $1 - \kappa^n_t$ and its data residue with probability κ^n_t. Differentiating the path gives the flux $\dot\kappa^n_t$ from the mask to the data residue. To recover the conditional jump rate, we divide this flux by the probability that the position is still masked:

$$u^i_t(a, z \mid x_1) = \frac{\dot\kappa^n_t}{1 - \kappa^n_t}\big[\delta(a, x^i_1) - \delta(a, z)\big] = \frac{\rho_n}{1-t}\big[\delta(a, x^i_1) - \delta(a, z)\big].$$

It depends on the pair only through the data residue, so averaging over the posterior replaces $\delta(a, x^i_1)$ with the denoiser:

$$u^i_t(a, x) = \frac{\rho_n}{1-t}\big[\,p^{\theta}_{1|t}(a \mid x) - \delta(a, x^i)\,\big].$$

Therefore, Option D. The bracket has nonnegative off-diagonal entries and sums to zero, as a valid velocity must. Once a position holds its data residue, its velocity is zero, so it retains that residue.

Option A drops the point mass on the current residue from both entries, so the rates no longer sum to zero. Option B reverses the bracket in the model velocity, which would give negative rates toward every other residue. Option C uses the backward factor $\dot\kappa^n_t/\kappa^n_t$, which belongs to the reverse

direction of 4(c). Option E uses the linear schedule's factor, which holds only when $\rho_n = 1$. Option F uses $\dot{\kappa}_t^n$ without dividing by the probability still on the source residue.

(b) To train the denoiser, we sample a time t and a pair (x_0, x_1). We use the conditional path to draw a noisy peptide x_t, then evaluate the cross-entropy of the predicted residue distribution at each position.

Which expression is that loss, written for the class-rate path? Choose ONE.

A. $-\mathbb{E}\Big[\sum_{i=1}^{N} \frac{\rho_{c(i)}}{1-t} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

B. $-\mathbb{E}\Big[\sum_{i:\, x_t^i = m} \frac{\rho_{c(i)}}{1-t} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

C. $-\mathbb{E}\Big[\sum_{i=1}^{N} \frac{1}{1-t} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

D. $-\mathbb{E}\Big[\sum_{i:\, x_t^i = m} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

E. $-\mathbb{E}\Big[\sum_{i=1}^{N} \rho_{c(i)}(1-t)^{\rho_{c(i)}-1} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

F. $-\mathbb{E}\Big[\sum_{i=1}^{N} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big]$

(0.5 points)

We train the denoiser by summing the cross-entropy of the clean residue predictions over all positions. Each position enters the loss with equal weight:

$$\mathcal{L}(\theta) = -\,\mathbb{E}_{t,(x_0,x_1),x_t}\Big[\sum_{i=1}^{N} \log p^{\theta}_{1|t}(x_1^i \mid x_t)\Big].$$

Therefore, Option F. The class rates determine how we draw the noisy peptide x_t. Once we have drawn it, we evaluate the cross-entropy without a schedule weight. At a visible position the true denoiser is a point mass on the residue already shown, so that position contributes nothing at the optimum. With no weight and no jump rate in the loss, we can choose the schedule and the sampler after training.

Option A weights each position by its class factor. Option B weights the masked positions by their class factor, which is the form that gives MDLM's likelihood bound. Option C applies the linear schedule's weight to every position. Option D keeps the masked positions and drops the weight. Option E weights by $\dot{\kappa}_t^n$. Every one of these has the same minimizer, since any positive weight is minimized by the true denoiser at every noisy peptide, and they differ in how much they emphasize each noise level during training.

(c) We can score one training peptide by hand.

For a four-position peptide, the network assigns the correct residues probabilities 1/2, 1/4, 1/4, and 1/2, respectively. What is the total cross-entropy across the four positions? Choose ONE.

A. $\frac{3}{2}\log 2$

B. $2\log 2$

C. $3\log 2$

D. $4\log 2$

E. $6\log 2$

F. $12\log 2$

(0.5 points)

The loss adds the negative log probability of the true data residue at each of the four positions:

$$-\log\tfrac{1}{2} - \log\tfrac{1}{4} - \log\tfrac{1}{4} - \log\tfrac{1}{2} = \log 2 + 2\log 2 + 2\log 2 + \log 2 = 6\log 2.$$

Therefore, Option E. The two positions at 1/4 contribute twice as much as the two at 1/2. A position contributes little to the loss when its neighbors make the clean residue easy to predict. A masked position contributes more when its neighbors leave the residue uncertain.

Option A averages over the four positions. Option B counts the two positions at 1/2 alone. Option C counts one position at 1/2 and one at 1/4. Option D uses 1/4 at two positions and 1 at the others. Option F doubles the sum, as if each position were counted twice.

(d) Several clean peptides can produce the same noisy peptide. This uncertainty limits how low the denoising loss can fall.

At its minimum over θ, what does the expected loss in (b) equal? Choose ONE.

A. The expected entropy of the true denoiser $p_{1|t}(\cdot \mid x_t)$, summed over the positions.

B. Zero, because the network can match the data residue at every position.

C. The expected KL divergence from the true denoiser to the network's prediction.

D. The entropy of the data distribution q.

E. The expected variance of the conditional velocity.

F. The expected cross-entropy between the source distribution and the data distribution.

(0.5 points)

Average over the data residue at a position before averaging over the noisy peptide. Given x_t, the clean residue follows the true posterior. Its expected loss is the cross-entropy between this posterior and the network prediction. We can separate this cross-entropy into two terms:

$$\mathbb{E}\big[-\log p^{\theta}_{1|t}(x_1^i \mid x_t)\big] = \mathbb{E}\Big[H\big(p_{1|t}(\cdot \mid x_t)\big)\Big] + \mathbb{E}\Big[D_{\mathrm{KL}}\big(p_{1|t}(\cdot \mid x_t) \,\|\, p^{\theta}_{1|t}(\cdot \mid x_t)\big)\Big].$$

Only the second term depends on θ. This divergence is nonnegative and equals zero when the network predicts the true posterior. Therefore, Option A. When several clean residues are consistent with the same noisy peptide, the posterior assigns probability to each of them. Its entropy measures the uncertainty that remains even with a perfect denoiser. In continuous flow matching, the matching quantity was the variance of the conditional velocity.

Option B would need the true denoiser to be a point mass at every noisy peptide. Option C is the part of the loss that the network drives to zero. Option D is the entropy of the data distribution, which is a different average and takes no account of the visible residues. Option E is the continuous analogue, which arises when the target is a vector and the loss is a squared error. Option F compares the source with the data, and the loss compares the data residue with a prediction made from the noisy peptide.

Step 4: Sampling, Correcting, and Checking Optimality (2 points)

Now that we have trained the denoiser, we can generate a peptide by starting from the all-mask source and taking Euler steps to $t = 1$. The schedule determines the probability of redrawing each position. We must choose the step length so that $\lambda_t \leq 1$.

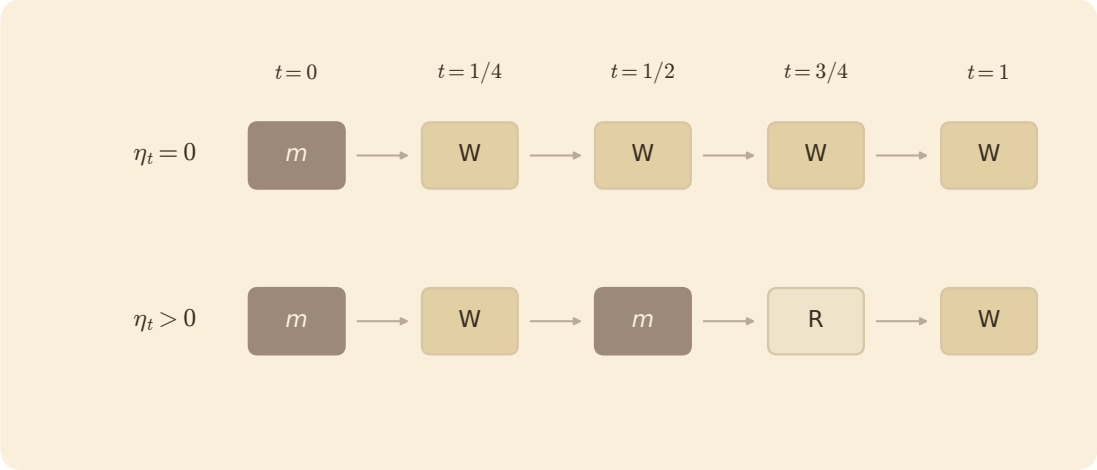


Figure 3. Two sample trajectories at one peptide position. Without a corrector, a revealed residue stays fixed. With a corrector, we can mask that position again and draw a new residue.

(a) At each step, we redraw a position from the denoiser with probability $\lambda_t = h\,\rho_n/(1-t)$ and keep its current residue otherwise.

With $h = 1/8$ at $t = 1/2$, which pair gives λ_t at an anchor position and at a scaffold position? Choose ONE.

A. $(1/2,\ 1/8)$

B. $(1,\ 1/4)$

C. $(1/4,\ 1)$

D. $(1/2,\ 1/2)$

E. $(2,\ 1/2)$

F. $(4,\ 1)$

(0.5 points)

The redraw probability is the step length times the schedule factor:

$$\lambda_t = h\,\frac{\rho_n}{1-t}.$$

At $t = 1/2$ with $h = 1/8$,

$$\text{anchor: } \tfrac{1}{8}\cdot\frac{4}{1/2} = 1, \qquad \text{scaffold: } \tfrac{1}{8}\cdot\frac{1}{1/2} = \tfrac{1}{4}.$$

Therefore, Option B. At this time, every masked anchor is redrawn. Each masked scaffold position has a one-in-four chance of being redrawn. The anchor value sits exactly at the limit, so a step any longer than 1/8 would ask for a probability above one at the anchors.

Option A halves the anchor value and uses h alone at the scaffold. Option C exchanges the two classes. Option D gives both classes the same value, which would hold only if $\rho_{\mathrm{A}} = \rho_{\mathrm{S}}$. Option E drops the factor h at the scaffold and doubles the anchor value. Option F is the pair of schedule factors $\rho_n/(1-t)$ before multiplying by h, divided by two.

(b) We take eight uniform steps from $t = 0$ to $t = 1$, so $h = 1/8$ and step k begins at $t = k/8$ for $k = 0, \ldots, 7$.

At which of these times does λ_t first rise strictly above one at an anchor position? Choose ONE.

A. $t = 1/8$

B. $t = 1/2$

C. $t = 5/8$

D. $t = 3/4$

E. $t = 7/8$

F. It stays at or below one at all eight times.

(0.5 points)

With $S = 8$ uniform steps, $h = 1/8$ and step k begins at $t = k/8$, so at an anchor position

$$\lambda_{t_k} = \frac{1}{8} \cdot \frac{4}{1 - k/8} = \frac{4}{8 - k}.$$

Reading down the grid gives 0.5, 4/7, 2/3, 0.8, 1, 4/3, 2, 4 at $k = 0, \ldots, 7$. The value first reaches one at $k = 4$, corresponding to $t = 1/2$. Since $\lambda_t = 1$ is a valid probability, the first invalid step is $k = 5$. Therefore, Option C. Three of the eight steps give redraw probabilities above one. Capping them at one introduces the sampling error we will examine as ρ_{A} increases in Step 6. We can keep $\lambda_t \le 1$ near $t = 1$ by shortening the step to $h \le (1 - \kappa_t)/\dot{\kappa}_t$, which here is $(1-t)/\rho_n$.

Option A is the second step, where the value is 4/7. Option B is where the value reaches one exactly, and a step with $\lambda_t = 1$ draws every masked anchor and stays valid. Option D and Option E are later steps, which also exceed one, and the question asks for the first. Option F would hold at a scaffold position, where $\lambda_{t_k} = 1/(8-k)$ reaches one only at the last step.

(c) To revise an early choice while preserving the probability path, we can add a corrector with zero net flux. The backward velocity at a position of class n is $\breve{u}_t^i(a, x) = \frac{\dot{\kappa}_t^n}{\kappa_t^n}\big[\delta(a, x^i) - p_{0|t}(a \mid x)\big]$. For the all-mask source, $p_{0|t}(\cdot \mid x) = \delta_m$.

Write d for $p_{1|t}^\theta(a \mid x)$, let $\hat{u}_t$ be the model velocity of 3(a), and take $\bar{u}_t = \hat{u}_t + \eta_t(\hat{u}_t - \breve{u}_t)$. Which pair gives the rate of revealing residue a at a masked position of class n and the rate of remasking a visible one? Choose ONE.

A. $\left(\frac{\rho_n}{1-t}\, d, \quad \eta_t \frac{\dot{\kappa}_t^n}{\kappa_t^n}\right)$

B. $\left((1+\eta_t)\frac{\rho_n}{1-t}\, d, \quad \eta_t \frac{\rho_n}{1-t}\right)$

C. $\left(\eta_t \frac{\rho_n}{1-t}\, d, \quad (1+\eta_t)\frac{\dot{\kappa}_t^n}{\kappa_t^n}\right)$

D. $\left((1+\eta_t)\frac{\rho_n}{1-t}\, d, \quad 0\right)$

E. $\left((1-\eta_t)\frac{\rho_n}{1-t}\, d, \quad \eta_t \frac{\dot{\kappa}_t^n}{\kappa_t^n}\right)$

F. $\left((1+\eta_t)\frac{\rho_n}{1-t}\, d, \quad \eta_t \frac{\dot{\kappa}_t^n}{\kappa_t^n}\right)$

(0.5 points)

To compute the corrected rates, we evaluate both velocities at the same time. At a masked position the backward velocity is zero toward a residue $a \neq m$, since $\delta(a, m) = 0$ and $p_{0|t}(a \mid x) = \delta_m(a) = 0$, so the combination keeps the forward rate alone and scales it:

$$\bar{u}_t^i(a, x) = (1+\eta_t)\,\frac{\rho_n}{1-t}\, d - \eta_t \cdot 0 = (1+\eta_t)\,\frac{\rho_n}{1-t}\, d.$$

At a visible position the forward velocity is zero toward the mask, since the denoiser places no mass there and the current residue is not the mask, while the backward velocity is $-\dot{\kappa}_t^n/\kappa_t^n$, so

$$\bar{u}_t^i(m, x) = 0 - \eta_t\Big(-\frac{\dot{\kappa}_t^n}{\kappa_t^n}\Big) = \eta_t\, \frac{\dot{\kappa}_t^n}{\kappa_t^n}.$$

Therefore, Option F. The extra term $\eta_t(\hat{u}_t - \breve{u}_t)$ satisfies detailed balance with p_t, since the flux it sends from the mask to a residue, $(1-\kappa_t^n)\eta_t \frac{\dot{\kappa}_t^n}{1-\kappa_t^n} d = \eta_t \dot{\kappa}_t^n d$, equals the flux it sends back, $\kappa_t^n d \cdot \eta_t \frac{\dot{\kappa}_t^n}{\kappa_t^n}$. The corrector contributes zero net flux at each state, so p_t follows the same path. Individual positions can now jump more often, allowing us to revise a residue chosen early.

Option A drops the factor $1+\eta_t$ from the reveal rate, which would move the path. Option B uses the forward factor for remasking, which breaks detailed balance except when $\kappa_t^n = 1 - \kappa_t^n$. Option C exchanges the two weights. Option D sets the remasking rate to zero, which leaves a reveal rate that is too large for the path. Option E subtracts η_t from the reveal rate, which would slow generation while the remasking stays.

(d) We can also choose a velocity by minimizing its kinetic energy along the conditional path. For the edge weight $w_t(y, x) = 1/p_t(y)$, the minimizing velocity is $u_t^\star(y, x \mid x_1) = p_t(y \mid x_1)\big[\partial_t \log p_t(y \mid x_1) - \partial_t \log p_t(x \mid x_1)\big]_+$. When the path depends on distances between residues, jumps occur only toward residues closer to the data residue.

Take the toy alphabet $\{\mathrm{L}, \mathrm{I}, \mathrm{K}, \mathrm{R}\}$ with $\Delta = 1$ inside the hydrophobic pair $\{\mathrm{L}, \mathrm{I}\}$ and inside the basic pair $\{\mathrm{K}, \mathrm{R}\}$, and $\Delta = 2$ across the pairs. The path is $p_t(y \mid x_1) \propto e^{-\beta_t \Delta(y, x_1)}$ with $\beta_t = t/(1-t)$. At $t = 1/2$ we have $\beta_t = 1$ and $\dot\beta_t = 4$, and with $x_1 = \mathrm{L}$ the path gives (0.6103, 0.2245, 0.0826, 0.0826) over $(\mathrm{L}, \mathrm{I}, \mathrm{K}, \mathrm{R})$. The current residue is K. Which rates does $u_t^\star(\cdot, \mathrm{K} \mid \mathrm{L})$ give toward $(\mathrm{L}, \mathrm{I}, \mathrm{R})$? Choose ONE.

A. (2.441, 0.449, 0)

B. (4.882, 0.898, 0)

C. (0, 0.898, 0.661)

D. (0.898, 4.882, 0)

E. (4.882, 1.796, 0)

F. (4.882, 0.898, 0.661)

(0.5 points)

At $t = 1/2$ the path is $p_t(\cdot \mid \mathrm{L}) = (0.6103, 0.2245, 0.0826, 0.0826)$. For the metric path, $\log p_t(y \mid x_1) = -\beta_t \Delta(y, x_1) + \text{const}$, so the two log-derivatives differ by $\dot\beta_t\big[\Delta(x, x_1) - \Delta(y, x_1)\big]$ and the kinetic-optimal velocity is

$$u_t^\star(y, x \mid x_1) = p_t(y \mid x_1)\, \dot\beta_t \big[\Delta(x, x_1) - \Delta(y, x_1)\big]_+.$$

The current residue K sits at distance $\Delta(\mathrm{K}, \mathrm{L}) = 2$, and $\dot\beta_t = 4$:

$$\mathrm{L}: \ 0.6103 \cdot 4 \cdot (2-0) = 4.882, \quad \mathrm{I}: \ 0.2245 \cdot 4 \cdot (2-1) = 0.898, \quad \mathrm{R}: \ 0.0826 \cdot 4 \cdot (2-2) = 0.$$

Therefore, $\boxed{\text{Option B.}}$ From K, a jump to L reduces the distance to the target by two, while a jump to I reduces it by one. A jump to R leaves the distance unchanged, so its rate is zero. For the mixture path in 3(a), minimizing the same energy gives the class-rate velocity. Each position then makes one jump from the mask to its clean residue.

Option A uses $\dot\beta_t = 2$. Option C weights each rate by the distance Δ to the data residue, which would forbid the move to L and allow the move to R. Option D exchanges L and I. Option E uses a distance reduction of two for the jump from K to I. The reduction is one. Option F assigns a positive rate to the jump to R even though the distance does not decrease.

Step 5: Relaxing the Peptide to the Simplex (2 points)

So far, each position has changed through discrete jumps. We can also represent a position by a probability vector and move it continuously toward a residue at a corner of the simplex. We then sample by integrating an ODE, which we can guide with a classifier or distill into a faster sampler. For the toy alphabet $\{\mathrm{L}, \mathrm{I}, \mathrm{K}, \mathrm{R}\}$, the simplex is a tetrahedron with corners $e_{\mathrm{L}}, e_{\mathrm{I}}, e_{\mathrm{K}}, e_{\mathrm{R}}$.

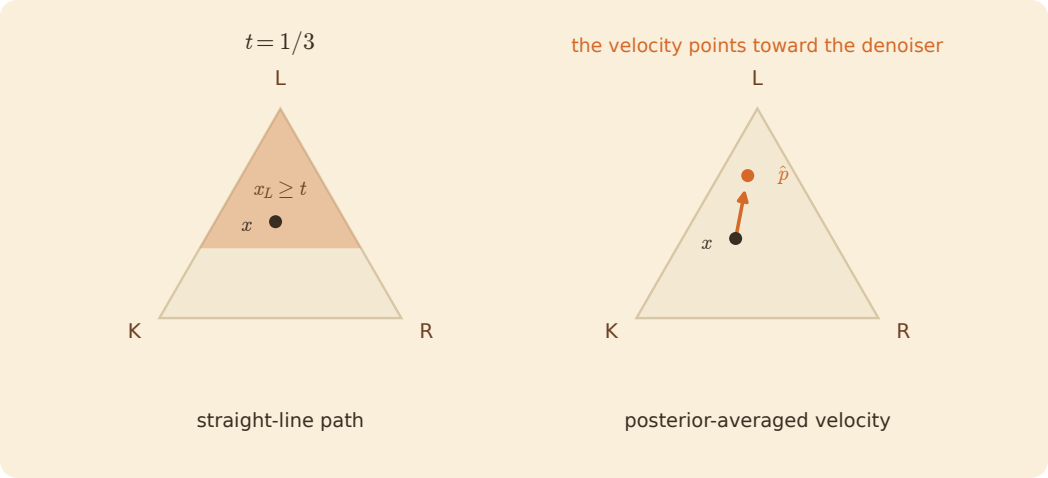


Figure 4. A triangle stands for the simplex over three of the four residues, with two separate points x. On the left, the points that could have come from the corner e_{L} under the straight-line path satisfy $x_{\mathrm{L}} \geq t$. On the right, the marginal velocity at a different point runs toward the denoiser $\hat{p}$.

(a) Consider the straight-line path $\psi_t = (1-t)x_0 + t\, e_k$ with x_0 drawn uniformly on the simplex. A point on a path ending at corner e_k must satisfy $x_k \geq t$. We can use this condition together with the fact that the coordinates sum to one.

At $t = 0.45$, at most how many of the four corners keep positive posterior probability at a position? Choose ONE.

A. 0

B. 1

C. 2

D. 3

E. It depends on the point x, so no bound holds at a fixed time.

F. All four, because every corner keeps positive probability at every finite time.

(0.5 points)

Suppose k corners still have positive posterior probability at time t. A point reachable from corner

e_j satisfies $x_j \geq t$, so each of those k coordinates is at least t, and the coordinates of a simplex point sum to one:

$$1 = \sum_j x_j \;\geq\; k\,t \qquad \Longrightarrow \qquad k \leq \frac{1}{t}.$$

At $t = 0.45$ this gives $k \leq 20/9$, so at most two corners remain possible. Two coordinates can reach the threshold because $0.45 + 0.45 \leq 1$. Therefore, Option C. The straight-line path rules the corners out one at a time, the first at $t = 1/K$, and the posterior becomes discontinuous in both time and space, which makes numerical integration sensitive to the step size. A Dirichlet path keeps every corner possible at finite time, since its density is positive wherever the data coordinate is positive.

Option A would need every coordinate to fall below t, and they sum to one. Option B holds only once t exceeds $1/2$, and $0.45 < 1/2$. Option D holds while t sits between $1/4$ and $1/3$. Option E misreads the bound, which follows from the sum constraint at the fixed time t and holds at every point of the simplex. Option F states the Dirichlet property, and the straight-line path here loses support.

(b) The marginal velocity averages the conditional velocities over the corners, weighted by the denoiser.

At $t = 1/2$ a position sits at $x = (0.40, 0.30, 0.20, 0.10)$ and the denoiser gives $\hat{p} = (0.50, 0.20, 0.20, 0.10)$ over $(\mathrm{L}, \mathrm{I}, \mathrm{K}, \mathrm{R})$. With the conditional velocity $(e_k - x)/(1-t)$ toward corner e_k, what is the marginal velocity at this position? Choose ONE.

A. $(0.20,\ -0.20,\ 0,\ 0)$

B. $(0.10,\ -0.10,\ 0,\ 0)$

C. $(0.40,\ -0.40,\ 0,\ 0)$

D. $(1.00,\ 0.40,\ 0.40,\ 0.20)$

E. $(0.20,\ 0.20,\ 0,\ 0)$

F. $(-0.20,\ 0.20,\ 0,\ 0)$

(0.5 points)

We average the conditional velocity over possible target corners using the denoiser probabilities. Only the target corner varies across terms, so this average gives

$$\sum_k \hat{p}_k \,\frac{e_k - x}{1-t} = \frac{1}{1-t}\Big(\sum_k \hat{p}_k e_k - x\Big) = \frac{\hat{p} - x}{1-t}.$$

With $\hat{p} - x = (0.10,\ -0.10,\ 0,\ 0)$ and $1 - t = 1/2$,

$$\frac{\hat{p} - x}{1-t} = (0.20,\ -0.20,\ 0,\ 0).$$

Therefore, Option A. The coordinates sum to zero, so the velocity is tangent to the simplex and the point stays on it. This is the same shape as the discrete velocity of 3(a), with the posterior mean of the one-hot data residue in place of the denoiser distribution and the current point in place of the point mass.

Option B drops the division by $1-t$. Option C divides by $1-t$ twice. Option D is $\hat{p}/(1-t)$, which omits the current point and does not sum to zero. Option E has both of the first two coordinates positive, so its entries sum to 0.4 and the point would leave the simplex. Option F reverses the direction, which would move the point away from the denoiser.

(c) To measure Fisher-Rao distance on the simplex, we first apply the square-root map to each probability vector. We can then compute the distance from the angle between the resulting points on the unit sphere.

What is the Fisher-Rao distance from the uniform point (1/4, 1/4, 1/4, 1/4) to the corner e_{L}? Choose ONE.

A. $\pi/3$

B. $\pi/2$

C. $2\pi/3$

D. $3\pi/4$

E. $2\arccos(1/4)$

F. π

(0.5 points)

The square-root map sends each probability vector to the unit sphere. To find the Fisher-Rao distance between two distributions, we double the angle between their images:

$$d_{\mathrm{FR}}(p,q) = 2\arccos\Big(\textstyle\sum_k \sqrt{p_k}\sqrt{q_k}\Big).$$

The uniform point maps to $(1/2, 1/2, 1/2, 1/2)$ and the corner e_{L} maps to $(1, 0, 0, 0)$, so the inner product is $1/2$ and

$$d_{\mathrm{FR}} = 2\arccos\tfrac{1}{2} = 2\cdot\frac{\pi}{3} = \frac{2\pi}{3}.$$

Therefore, Option C. Under this metric, a given change in probability contributes more to distance when the residue is rare. Any two distinct corners have distance π.

Option A is the angle itself, without the factor of two. Option B is the distance that two orthogonal square-root vectors would give before doubling. Option D is a value between the angle and the distance, and the arccosine of $1/2$ fixes the answer at $2\pi/3$. Option E uses the probability $1/4$ in place of its square root inside the arccosine. Option F is the distance between two distinct corners, where the inner product is zero.

(d) The network can predict a velocity with a component normal to the sphere. To move along the sphere, we subtract that component.

At the sphere point $x = (1/2,\ 1/2,\ 1/2,\ 1/2)$ the network outputs $v = (0.4,\ 0.2,\ -0.2,\ 0.0)$. Which velocity do we use? Choose ONE.

A. $(0.40,\ 0.20,\ -0.20,\ 0.00)$

B. $(0.30,\ 0.10,\ -0.30,\ -0.10)$

C. $(0.50,\ 0.30,\ -0.10,\ 0.10)$

D. $(0.20,\ 0.00,\ -0.40,\ -0.20)$

E. $(0.10,\ -0.10,\ -0.50,\ -0.30)$

F. $(0.30,\ 0.10,\ -0.30,\ 0.10)$

(0.5 points)

We keep the part of the output that is tangent to the sphere and subtract the part along the current point:

$$v_{\tan} = v - \langle x, v\rangle\, x.$$

Here $\langle x, v\rangle = \frac{1}{2}(0.4 + 0.2 - 0.2 + 0.0) = 0.2$, so $\langle x, v\rangle x = (0.1,\ 0.1,\ 0.1,\ 0.1)$ and

$$v_{\tan} = (0.4,\ 0.2,\ -0.2,\ 0.0) - (0.1,\ 0.1,\ 0.1,\ 0.1) = (0.3,\ 0.1,\ -0.3,\ -0.1).$$

Therefore, $\boxed{\text{Option B.}}$ The result has zero inner product with x, so the projected velocity is tangent to the sphere. We can apply this projection to the output of any network architecture.

Option A is the raw output, which points partly away from the sphere. Option C adds the radial part in place of subtracting it. Option D subtracts twice the radial part, which reflects the output through the tangent plane. Option E subtracts three times the radial part. Option F changes the sign of the last entry, so its inner product with x is 0.1 and the point would drift off the sphere.

Step 6: Guiding Generation and Reading the Results (2 points)

So far, we have generated peptides without controlling whether they bind the target. To guide generation toward binders, we can keep the trained denoiser and condition on a binding event $\mathcal{B}$ using Doob's h-transform. Here $\mathcal{B}$ is the event that a binding classifier accepts the finished peptide. Given the current peptide, the probability of this event is $h_t(x) = \Pr(\mathcal{B} \mid x_t = x)$.

(a) The value $h_t(x)$ is the probability of the event at the end of generation, starting from the current peptide.

Write d for $p^{\theta}_{1|t}(a \mid x)$. Which expression is the guided rate of revealing residue a at a masked position of class n? Choose ONE.

A. $\dfrac{\rho_n}{1-t}\, d\, \dfrac{h_t(x)}{h_t(x^{i \leftarrow a})}$

B. $\dfrac{\rho_n}{1-t}\, d\, h_t(x^{i \leftarrow a})$

C. $\dfrac{\rho_n}{1-t}\, d\, \dfrac{h_t(x^{i \leftarrow a})}{h_t(x)}$

D. $\dfrac{\dot{\kappa}^n_t}{\kappa^n_t}\, d\, \dfrac{h_t(x^{i \leftarrow a})}{h_t(x)}$

E. $\dfrac{\rho_n}{1-t}\, d + h_t(x^{i \leftarrow a}) - h_t(x)$

F. $\dfrac{\rho_n}{1-t}\, d\, \dfrac{h_t(x^{i \leftarrow a})}{1 - h_t(x)}$

(0.5 points)

Doob's h-transform multiplies each rate by the value at the destination divided by the value at the origin, $u^h_t(y, x) = u_t(y, x) h_t(y)/h_t(x)$. The unguided rate of revealing a at a masked position of class n is $\frac{\rho_n}{1-t} d$, its origin is x, and its destination is $x^{i \leftarrow a}$, so

$$u^{\mathcal{B}}_t\big(x^{i \leftarrow a}, x\big) = \frac{\rho_n}{1-t}\, d\, \frac{h_t(x^{i \leftarrow a})}{h_t(x)}.$$

Therefore, Option C. The value h_t is harmonic for the unguided process, so the transformed process is the model conditioned on the event, exact in the continuous-time limit and with the true classifier. With the denoiser fixed, guidance increases the rates toward residues that make the binding event more likely and decreases the rates toward residues that make it less likely.

Option A inverts the ratio, which would steer away from binders. Option B drops the division by $h_t(x)$, so every rate would shrink with the overall value of the current peptide. Option D uses the backward factor from 4(c) in place of the reveal rate. Option E adds the change in value to the rate, and the h-transform multiplies by a ratio of values. Option F divides by the probability that the event fails.

(b) A classifier on noisy peptides can be hard to train, so instead, we can estimate the value by completing the peptide with the model and scoring each completion. To adjust the strength of guidance, we raise the value ratio to a power γ before multiplying it by the unguided rate.

At a masked anchor position at $t = 1/2$, the unguided rates toward L and K are 4 and 2. Five completions after revealing L earn rewards $(1, 1, 0, 1, 1)$, and five after revealing K earn $(0, 1, 0, 0, 0)$. The current value estimate is $1/2$, and we take $\gamma = 2$. Given that the next reveal at this position is L or K, what is the probability that it is L? Choose ONE.

A. 2/3

B. 4/5

C. 8/9

D. 16/17

E. 32/33

F. 64/65

(0.5 points)

Each value estimate is the average reward over the completions:

$$h(\mathrm{L}) = \frac{1+1+0+1+1}{5} = \frac{4}{5}, \qquad h(\mathrm{K}) = \frac{0+1+0+0+0}{5} = \frac{1}{5}.$$

With $\gamma = 2$ we raise each value ratio to the second power before multiplying the unguided rate:

$$\mathrm{L}: \ 4\left(\frac{4/5}{1/2}\right)^2 = 4 \cdot \frac{64}{25} = \frac{256}{25}, \qquad \mathrm{K}: \ 2\left(\frac{1/5}{1/2}\right)^2 = 2 \cdot \frac{4}{25} = \frac{8}{25}.$$

Given that the next reveal is one of the two, its probabilities are proportional to these rates:

$$\frac{256/25}{256/25 + 8/25} = \frac{256}{264} = \frac{32}{33}.$$

Therefore, $\boxed{\text{Option E.}}$ The current value of $1/2$ sets the size of the two guided rates, and it cancels from their ratio at every γ. The exact conditional sampler uses $\gamma = 1$. A higher power of h generally loses the harmonic property of the original value function. In this example, increasing the power further favors L over K.

Option A is the unguided probability, $4/(4+2)$. Option B compares the two values without the base rates. Option C is the answer at $\gamma = 1$, where the rates are $32/5$ and $4/5$. Option D would follow from rates in the ratio 16 to 1. Option F would follow from $\gamma = 3$ at these values, where the preference grows again.

Suppose we train class-rate flow matching on a set of peptide binders. We vary ρ_{A} while holding $\rho_{\mathrm{S}} = 1$ fixed. We will use the hypothetical results in Table 1 and Figure 5 to examine how the anchor schedule affects recovery at two sampling budgets.

Model	ρ_{A}	8 steps	128 steps
Autoregressive	–	–	61.4
Class-rate (uniform)	1	42.3	60.9
Class-rate	2	48.1	61.1
Class-rate	4	55.6	61.2
Class-rate	8	52.0	60.6
Class-rate	16	44.7	58.9

Table 1. Hypothetical anchor recovery (%) on held-out binders, at 8 and at 128 uniform sampling steps. The autoregressive model takes one network call per residue, so it costs 24 calls and has no eight-step entry.

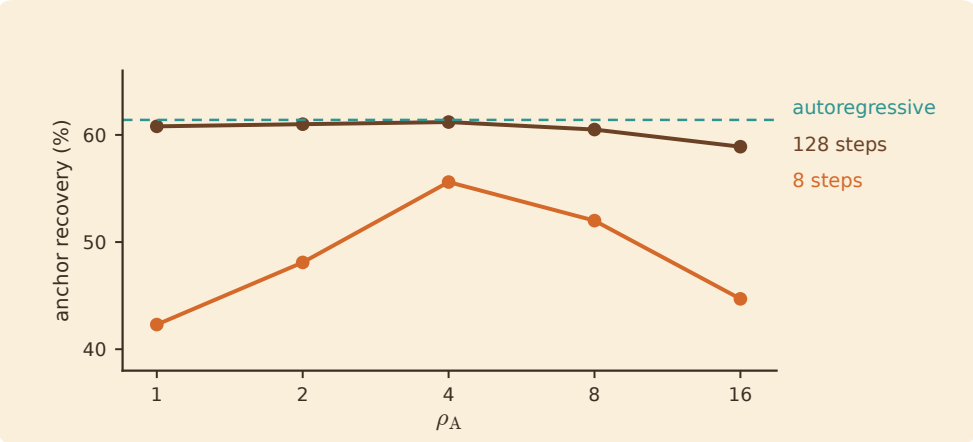


Figure 5. The values of Table 1 against ρ_{A} on a logarithmic axis. The dashed line marks the autoregressive model.

(c) With eight sampling steps, recovery rises from $\rho_A = 1$ to $\rho_A = 4$ before falling at larger exponents.

How can we explain this trend given the eight-step sampling budget? Choose ONE.

A. Larger ρ_A reveals the anchors earlier and helps at every value, so the fall from 55.6 to 44.7 after $\rho_A = 4$ reflects noise in the evaluation.

B. Larger ρ_A lowers λ_t at the anchors, so fewer of the eight steps are clipped and recovery rises across the whole sweep.

C. Larger ρ_A raises the loss weight at the anchors, so the denoiser fits them at the expense of the scaffold.

D. Revealing anchors earlier helps us predict scaffold residues. Beyond $\rho_A = 4$, more of the eight steps have $\lambda_t > 1$, so the cost of clipping exceeds the benefit of revealing anchors early.

E. Larger ρ_A needs more network calls per step at the anchor positions, so the eight-step budget buys fewer reveals there and recovery falls once the anchors outpace it.

F. Recovery at eight steps must sit below recovery at 128 steps, so the eight-step column follows the 128-step column and nothing about ρ_A follows.

(0.5 points)

Two effects run against each other as ρ_A grows. A larger ρ_A reveals the anchors earlier, so the scaffold positions are drawn with the motif already in place and the parallel draw loses less of the dependence measured in 1(d). A larger ρ_A also raises $\lambda_{t_k} = \rho_A/(8-k)$ at every step, and a redraw probability above one has to be capped, which moves the sampler off the path. Counting the steps of the eight-step grid with $\lambda_{t_k} > 1$ gives

$$\rho_A = 1:\ 0, \qquad 2:\ 1, \qquad 4:\ 3, \qquad 8:\ 7, \qquad 16:\ 8.$$

The recovery peaks where the two effects cross, at $\rho_A = 4$. Therefore, **Option D.** At $\rho_A = 8$ seven of the eight steps are capped, and at $\rho_A = 16$ every step is, so the sampler no longer follows the path it was trained for. We would choose ρ_A using the same step budget we plan to use for generation.

Option A contradicts the column, which falls by almost 11 points from $\rho_A = 4$ to $\rho_A = 16$, so the reported recovery does not increase throughout the sweep. Option B has the inequality backward, since $\lambda_{t_k} = \rho_A/(8-k)$ rises with ρ_A at every step, so more steps are clipped and not fewer. Option C attributes the trend to a loss weight. In 3(b), each position contributes equally to the cross-entropy. Option E is false, because one step costs one network call at every ρ_A, and all 24 positions update in that call. Option F compares the two columns and says nothing about the shape of either, and the eight-step column rises and falls while the other stays nearly flat.

(d) With 128 sampling steps, anchor recovery stays near 61 percent for every ρ_{A} up to 8.

Which reading of that column agrees with the construction? Choose ONE.

A. The class rates cancel from the loss, so every value of ρ_{A} trains the same denoiser, and the 8-step and 128-step columns must therefore show the same values.

B. At 128 steps the redraw probability λ_{t_k} stays at or below one for every ρ_{A} in the table, so no step is clipped and each sampler follows its own path exactly.

C. The denoiser reaches zero loss at 128 steps, so recovery is limited by the data distribution alone.

D. The paths share a source at $t = 0$ and differ in where they end at $t = 1$, so the models agree early in generation and a long step budget hides the difference between their endpoints.

E. Every schedule has $\kappa_0^n = 0$ and $\kappa_1^n = 1$, so all paths end at the data distribution. With 128 steps, the sampling error is small.

F. Anchor recovery saturates at 61.4, so no model can be separated from the autoregressive one at any step count.

(0.5 points)

Every class schedule satisfies $\kappa_0^n = 0$ and $\kappa_1^n = 1$, so each of these paths starts at the all-mask peptide and ends at the data distribution, whatever ρ_{A} is. The schedules differ in when the anchors become visible. As we shorten the Euler steps, the sampler more closely follows the path specified by each schedule. Therefore, Option E. With 128 steps the models land within half a point of each other up to $\rho_{\mathrm{A}} = 8$. The fall to 58.9 at $\rho_{\mathrm{A}} = 16$ comes from the same capping as in 6(c), since $\lambda_{t_k} = 16/(128 - k)$ still exceeds one over the last fifteen steps, and those steps carry the residues that are hardest to place.

Option A overlooks how we obtain the training examples. The class rates determine the corruption used to draw x_t, so changing ρ_{A} changes the inputs used to train the denoiser. Its conclusion also fails on the table, where the two columns differ by 13 points at $\rho_{\mathrm{A}} = 1$. Option B claims no step is capped at 128 steps, and the last steps are capped for every $\rho_{\mathrm{A}} > 1$. Option C claims zero loss, and 3(d) put the minimum at the entropy of the true denoiser. Option D has the endpoints the wrong way around, since these paths share both the source and the data distribution. Option F reads a ceiling into the autoregressive value, and the eight-step column separates the models by more than 13 points.

End of practice exam.

This page is left intentionally blank for scratch work